

# Centralized Infrastructure Monitoring & Observability Platform

This document summarizes the current implementation status, completed work, roadmap, and target architecture for the centralized monitoring platform.

## Completed Work

Ubuntu Server 24.04 deployed and hardened; Docker Engine and Docker Compose installed; Docker networking and persistent volumes configured; Prometheus, Grafana, Loki, Alertmanager, Blackbox Exporter, and SNMP Exporter deployed successfully. The monitoring platform is operational and ready for onboarding infrastructure.

## Project Roadmap

### Phase 1 - Infrastructure Foundation [Completed]

- Ubuntu Server
- Server Hardening
- Docker Engine & Compose
- Docker Network & Volumes

### Phase 2 - Core Monitoring Platform [Completed]

- Prometheus
- Grafana
- Loki
- Alertmanager
- Blackbox Exporter
- SNMP Exporter

### Phase 3 - Infrastructure Monitoring

- Monitoring Server
- Proxmox
- Virtual Machines
- Dell iDRAC

### Phase 4 - Network Monitoring

- pfSense
- Cisco Switches
- HP Switches
- Internet Links

### Phase 5 - Application Monitoring

- NodeJS
- Python
- PHP
- Docker Containers

### Phase 6 - Database Monitoring

- MySQL
- PostgreSQL
- Redis

### Phase 7 - Website & Service Monitoring

- HTTP/HTTPS
- SSL
- DNS
- Ping

### Phase 8 - Centralized Logging

- Grafana Alloy
- System & Application Logs

### Phase 9 - Dashboards

- Executive
- NOC
- Infrastructure

- Network
- Database
- Application

#### Phase 10 - Alerting

- Email
- Telegram
- Slack

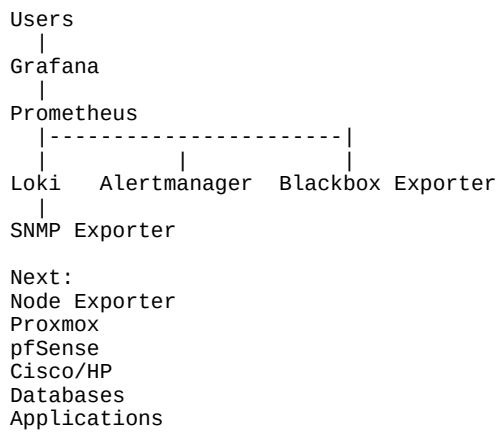
#### Phase 11 - Security & Operations

- HTTPS
- Nginx
- Backups
- Disaster Recovery

#### Phase 12 - Future Enhancements

- NetFlow
- SIEM
- CMDB
- Predictive Analytics

## Current Architecture



## Overall Progress

Infrastructure foundation and core monitoring platform are complete. The next milestone is onboarding production infrastructure, followed by dashboards, alerting, and centralized logging.